

DATORIX DATA VIGILANCE

Enterprise Database Activity Monitoring Platform

Continuous Agentless Monitoring, Deep Packet Inspection & Real-Time Lifecycle Protection

0ms	99.99%	100%	5+
ADDED LATENCY	UPTIME SLA	AGENTLESS FOOTPRINT	CORE ARCHITECTURES

PRODUCT OVERVIEW

The enterprise database represents an organization’s single largest unmonitored attack surface. Standard application-level security mechanisms frequently fail to guard against compromised service credentials, lateral movement, or direct database connections. Without wire-level visibility, malicious database manipulation and data exfiltration can occur completely undetected.

The **Datorix Enterprise Database Security Platform** eliminates these enterprise blind spots. Moving beyond traditional passive monitoring, Datorix combines non-intrusive network-level Deep Packet Inspection (DPI) with an active, inline Database Firewall. By parsing underlying database transaction engines without triggers or architectural modifications, Datorix records every login, query, schema modification, and administrative operation instantly, delivering absolute data visibility with zero performance impact on core production assets.

1. ADVANCED DATABASE FIREWALL & SECURITY POLICIES

The Datorix Database Firewall provides robust inline processing power capable of dropping active security threats, applying query access controls, and stopping policy infractions mid-flight. Utilizing deep SQL protocol inspection, the platform parses incoming wire traffic to validate structural integrity and isolate high-risk behavior before it reaches the data tier.

Technical Firewall Capabilities

- **Deep SQL Protocol Inspection:** Inspects database network protocols at the packet level to enforce strict protocol conformance, blocking malformed requests, buffer overflow attempts, and unauthorized protocol extensions.
- **SQL Grammar and Syntax Analysis:** Employs a built-in SQL parsing engine that deconstructs statements into abstract syntax trees (ASTs), checking structural integrity rather than relying solely on easily bypassed regular expressions.
- **Real-Time Query Inspection and Blocking:** Evaluates every incoming query against active security profiles, terminating violating connections or blocking individual statements mid-stream.
- **SQL Fingerprinting:** Generates unique cryptographic signatures for authorized transactional structures to classify safe queries versus novel, unexpected variations.
- **Query Whitelisting and Blacklisting:** Allows security administrators to establish explicit whitelists for trusted application query structures and blacklists for banned commands or administrative keywords.
- **Virtual Patching:** Shields known database vulnerabilities from exploitation by detecting and blocking specific targeted attack vectors at the network layer, removing the immediate pressure for vendor patch deployments.
- **Session Monitoring and Termination:** Tracks active database connections in real time, enabling automatic or manual termination of sessions exhibiting suspicious activity, brute-force patterns, or unauthorized connection pooling.
- **Privileged User Activity Monitoring:** Maintains strict surveillance over database administrators (DBAs), service accounts, and root users, alerting on and restricting schema modifications during unauthorized windows.
- **Context-Aware Policy Enforcement:** Couples query structure with external metadata—such as source client IP addresses, calling applications, and verified user identities.

- **Adaptive Risk-Based Access Control:** Dynamically adjusts access privileges based on shifting user risk scores, stepping up authentication requirements or blocking queries if behavioral baselines deviate.
- **Data Exfiltration Prevention:** Monitors query payload responses and row counts, terminating connections when an anomalous volume of sensitive data is requested.
- **Real-Time Anomaly Detection:** Leverages behavioral baselines to identify runtime anomalies, such as out-of-hours access, unusual data volume transfers, or unprecedented query execution speeds.

Data Governance & Compliance Enforcement

The Database Firewall serves as a programmatic enforcement mechanism for corporate governance frameworks, technical security standards, and international regulations (including SOX, HIPAA, GDPR, and PCI-DSS).

- **Role-Based Access Control (RBAC):** Synchronizes with enterprise identity providers to map organizational roles to granular query permissions, abstracting policy definition away from local database configurations.
- **Least Privilege Enforcement:** Restricts client applications and administrative accounts strictly to the minimal functional query footprint required for documented operations.
- **Zero Trust Database Security:** Mandates strict continuous verification for every database transaction, operating under the principle that no network location or application credential is inherently trusted.
- **Separation of Duties (SoD):** Prevents individual users from possessing overlapping administrative privileges, ensuring that developers, DBAs, and security auditors maintain distinct and restricted operational spheres.
- **Data Access Policies:** Enforces strict limitations on who can interact with rows containing Personally Identifiable Information (PII), Payment Card Industry (PCI) data, or Protected Health Information (PHI).
- **Business-Hour Access Restrictions:** Restricts administrative access or bulk exports to predefined operational windows, flagging or dropping connections initiated during off-hours.
- **Geo-Location-Based Access Controls:** Coordinates client IP attributes with geographical databases to block database traffic originating from unapproved regions or unexpected network domains.
- **Database Change Control Policies:** Records and regulates schema alterations, index drops, and user creation, matching modifications against approved change-management windows.
- **Audit and Compliance Enforcement:** Provides immutable, signed telemetry to verify to external auditors that security controls are being actively and uniformly enforced.

2. COMPREHENSIVE SQL INJECTION (SQLI) PROTECTION

SQL injection remains one of the most devastating methodologies for database exploitation and unauthorized data extraction. The Datorix platform features a multi-tiered defense engine designed specifically to intercept and neutralize structured threats before they manipulate the underlying database engine.

Attack Vectors Covered

- **Classic SQL Injection:** Detects and blocks the injection of simple malicious payloads into user input fields aimed at altering standard logical checks (e.g., ' OR '1'='1').
- **Union-Based SQL Injection:** Blocks the abuse of the UNION operator intended to combine the results of an application query with internal system metadata tables.
- **Blind SQL Injection:** Identifies sophisticated probing patterns where attackers attempt to infer database details by asking true/false questions, even when application error messages are suppressed.
- **Boolean-Based SQL Injection:** Neutralizes queries altered to evaluate conditional statements, which would
- **Error-Based SQL Injection:** Halts input structured to intentionally force database errors that would reveal schema details, table names, or software versions.
- **Second-Order SQL Injection:** Tracks and stops malicious payloads that are successfully stored in a database but later executed when processed by a separate, downstream application query.
- **Stored Procedure Injection:** Mitigates risks associated with parameters passed maliciously into backend stored procedures, preventing unintended execution of administrative commands.
- **Dynamic SQL Abuse:** Identifies instances within applications where user inputs are concatenated

otherwise alter application behavior based on the validity of an injected expression.

- **Time-Based SQL Injection:** Detects and thwarts payloads containing database sleep commands or heavy processing loops designed to infer information through network response latency.

directly into execution strings, ensuring structural safety through wire-level extraction.

- **Privilege Escalation Attempts:** Immediately intercepts query modifications intended to grant standard users elevated permissions or administrative roles.

Multi-Layered Protection Mechanisms

Rather than relying on signature matching alone, Datorix applies six distinct layers of security evaluation sequentially to all network database streams:

1. **SQL Parsing:** Deconstructs incoming database strings into tokenized syntax maps to isolate injected commands from legitimate query fields.
2. **Behavioral Analysis:** Monitors historical user baselines to spot anomalous structural shifts or unexpected deviations from standard query volume and complexity.
3. **Anomaly Detection:** Applies localized context checks to discover timing irregularities or out-of-bounds parameter lengths.
4. **Query Fingerprinting:** Utilizes pre-compiled, cryptographically validated templates of acceptable application behavior to quickly reject unrecognized or rogue structures.
5. **Contextual Validation:** Evaluates whether a query aligns with the verified user identity, calling application context, and historical client IP behaviors.
6. **Adaptive Learning & Real-Time Blocking:** Continuously refines security baselines to recognize evolving application updates while maintaining the ability to trigger immediate TCP resets to block threats mid-flight.

3. AUTOMATED DATABASE DISCOVERY & ASSET INVENTORY

An organization cannot secure assets it does not know exist. Datorix eliminates data visibility blind spots through an integrated, automated Database Discovery engine that acts continuously across the enterprise landscape.

- **Active and Passive Discovery:** Combines passive network listening via hardware TAPs/SPAN ports with scheduled active subnet scanning to catch both active data pools and rogue database instances.
- **Instance Identification:** Automatically fingerprints and classifies instances across multiple vendor types (MySQL, PostgreSQL, Oracle, SQL Server), identifying specific version numbers, patch levels, and port mappings.
- **Configuration Inventory:** Extracts and maintains a ledger of database operational settings, security parameters, user access rules, and network binding structures.
- **Sensitive Data Detection (Classification):** Leverages customizable discovery patterns to scan tables for sensitive data structures, automatically classifying regions housing PII, PCI data, PHI, financial ledgers, or intellectual property.
- **Continuously Updated Database Inventory:** Maintains a dynamic, real-time catalog of the entire database estate. When a new database instance spins up, it is immediately inventoried, analyzed, and onboarded into active security scanning workflows.

4. ENTERPRISE SIEM & SOC INTEGRATION

The Datorix platform is engineered to integrate seamlessly into modern enterprise Security Operations Centers (SOCs), transforming raw database events into contextualized, actionable security intelligence.

- **Centralized Monitoring & Incident Correlation:** Instead of siloing database telemetry, Datorix feeds structural security events upstream to existing Security Information and Event Management (SIEM) solutions (such as Splunk, Microsoft Sentinel, IBM QRadar, and Elastic Security). By emitting streaming webhooks and structured syslogs, the platform pairs database events with broader network, endpoint, and identity alerts.
- **Real-Time Threat Detection:** Converts wire-level database threats instantly into actionable security incidents, ensuring that security analysts see events the moment a policy-breaking action occurs.

- **Automated Response Workflows:** Integrates with Security Orchestration, Automation, and Response (SOAR) playbooks. An alert from Datorix can trigger automated responses, such as revoking a compromised user's active directory privileges or modifying firewall rules to block a rogue source IP address.

5. SYSTEM ARCHITECTURE: THE 5-STEP PIPELINE

The core architecture uses a non-intrusive network-level pipeline to process traffic with absolute isolation from active production environments.

01 CAPTURE DPI captures all network database traffic passively at the packet level. No local database installation.	02 ANALYZE Queries are fully parsed, isolated, and enriched with identity and contextual variables.	03 STORE Events are written down to cryptographically secure, SHA-256 protected immutable history blocks.	04 ALERT Webhook, SMS, and email alerts trigger on defined risk profiles or severe runtime anomalies.	05 REPORT One-click visual dashboard export of PDF, JSON, or CSV compliance templates for external auditors.
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6. TECHNICAL SPECIFICATIONS & INTEGRATION STACK

Datorix leverages highly efficient Change Data Capture (CDC) streaming pipelines to parse underlying transaction logs without triggers or schema changes, minimizing overhead and eliminating gaps.

SUPPORTED DATABASE	NATIVE CAPTURE ENGINE	DEPLOYMENT IMPACT
MySQL	Binlog Extraction	Zero-load transactional log tailing
PostgreSQL	Write-Ahead Log (WAL)	Zero-load transactional log tailing
Oracle	Redo Log Streams	Zero-load transactional log tailing
Microsoft SQL Server	Native CDC Pipelines	Zero-load transactional log tailing

Flexible Deployment Architectures

- **Guardian Proxy Mode:** Provides inline processing power capable of dropping active security threats, applying query firewall access controls, and stopping policy infractions mid-flight.
- **Watcher Sniffer Mode:** Operates strictly out-of-band using passive network hardware TAP or SPAN port aggregators to trace production infrastructure without contributing any risk layer or latency.

Beta Access & Demonstration Notice: Datorix is currently accepting enterprise registrants for its Early Access Beta Program. No credit card or cloud architectural overhaul is required to start a evaluation. Request a demonstration today.

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